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ELITE IAL MATHEMATICS

EDEXCEL INTERNATIONAL A LEVEL

WMA12/01 May/June 2024 Worked Solutions

May/June 2024 paper rebuilt with the worked solution after each question.

10
paper questions

WMA12
Pure 2

Single-paper solutions
standalone IAL route

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P2

PAST PAPER

WMA12/01 May/June 2024

May/June 2024 | 10 questions | 75 marks

10 questions	**75** marks	**Answers** worked solution after each question

WMA12/01 MAY/JUNE 2024

5 marks

Question 1

Binomial Expansion

1. (a) Find the first four terms, in ascending powers of x, of the binomial expansion of

$$\left(1-\frac{1}{6}x\right)^9$$

giving each term in simplest form.

(3)

(b) Hence find the coefficient of x^3 in the expansion of

$$(10x+3)\left(1-\frac{1}{6}x\right)^9$$

giving the answer in simplest form.

(2)

(Total for Question 1 is 5 marks)

(Total for Question 1 is 5 marks)

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Worked Solution - Question 1

Binomial Expansion

Topic group

1. Choose the required binomial terms

Write the expansion term-by-term and keep only the powers needed for the coefficient or approximation.

2. Compare coefficients

Collect like powers, compare with the given expression, and solve for the required constant.

3. State the final result

Write the answer in the form requested by the question: 7 53 10 1 3 18 6 \times+ \times- =

Final answer

7 53 10 1 3 18 6 \times+ \times- =

Question 2

7 marks

Arithmetic Sequences

2. **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

In an arithmetic series,

- the sixth term is 2
- the sum of the first ten terms is –80

For this series,

(a) find the value of the first term and the value of the common difference.

(4)

(b) Hence find the smallest value of n for which

$$S_n > 8000$$

(3)

(Total for Question 2 is 7 marks)

(Total for Question 2 is 7 marks)

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Worked Solution - Question 2

Arithmetic Sequences

Topic group

1. Use the arithmetic formula

Identify the first term, common difference, and required term or sum from the question.

2. Substitute and solve

Use the arithmetic term or sum formula, then simplify the requested value.

3. State the final result

Write the answer in the form requested by the question: 35 n =

Final answer

35 n =

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6 marks

Question 3

Laws of Logarithms

3. **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

(i) Using the laws of logarithms, solve

$$2\log_2(2-x) = 4 + \log_2(x+10)$$

(5)

(ii) Find the value of

$$\log_{\sqrt{a}} a^6$$

where a is a positive constant greater than 1

(1)

(Total for Question 3 is 6 marks)

(Total for Question 3 is 6 marks)

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Worked Solution - Question 3

Laws of Logarithms

Topic group

1. Combine the logarithms

Use the power, product, and quotient laws to rewrite the equation as a single logarithmic or exponential statement.

2. Solve with domain checks

Solve the resulting equation and reject any value that does not satisfy the original logarithms.

3. State the final result

Write the answer in the form requested by the question: The required value or expression is obtained by the final simplification.

Final answer
The required value or expression is obtained by the final simplification.

Question 4

8 marks

Polynomials

4. $$f(x) = (x - 2)(2x^2 + 5x + k) + 21$$

where k is a constant.

(a) State the remainder when $f(x)$ is divided by $(x - 2)$

(1)

Given that $(2x - 1)$ is a factor of $f(x)$

(b) show that $k = 11$

(2)

(c) Hence

(i) fully factorise $f(x)$,

(ii) find the number of real solutions of the equation

$$f(x) = 0$$

giving a reason for your answer.

(5)

(Total for Question 4 is 8 marks)

(Total for Question 4 is 8 marks)

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Worked Solution - Question 4

Polynomials

Topic group

1. Use the polynomial condition

Apply the factor theorem, remainder theorem, or substitution stated in the question to form the required equation.

2. Finish the algebra

Solve the resulting equation and substitute back where needed, then write the requested factor, coefficient, or value.

3. State the final result

Write the answer in the form requested by the question: has 1 root (at 1 2 x =)

Final answer

has 1 root (at 1 2 x =)

Question 5

6 marks

Proof

5. **In this question you must show detailed reasoning.**

(a) Given that x and y are positive numbers such that

$$(x-y)^3 > x^3 - y^3$$

prove that

$$y > x$$

(4)

(b) Using a counter example, show that the result in part (a) is not true for all real numbers.

(2)

(Total for Question 5 is 6 marks)

(Total for Question 5 is 6 marks)

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Worked Solution - Question 5

Proof

Topic group

1. Set up the proof

Translate the statement into algebra, or choose a clear counterexample if the statement is false. Keep each implication justified.

2. Complete the argument

Simplify the algebra or evaluate the counterexample, then finish with a clear conclusion.

3. State the final result

Write the answer in the form requested by the question: The statement is false; a valid counterexample completes the disproof.

Final answer
The statement is false; a valid counterexample completes the disproof.

Question 6

9 marks

Integration

6. (a) Sketch the curve with equation

$$y = a^x + 4$$

where a is a positive constant greater than 1

On your sketch, show

- the coordinates of the point of intersection of the curve with the y-axis
- the equation of the asymptote of the curve

(3)

x	2	2.3	2.6	2.9	3.2	3.5
y	0	0.3246	0.8629	1.6643	2.7896	4.3137

The table shows corresponding values of x and y for

$$y = 2^x - 2x$$

with the values of y given to 4 decimal places as appropriate.

Using the trapezium rule with all the values of y in the given table,

(b) obtain an estimate for $\int_2^{3.5} \left(2^x - 2x\right) \mathrm{d}x$, giving your answer to 2 decimal places.

(3)

(c) Using your answer to part (b) and making your method clear, estimate

(i) $\int_2^{3.5} \left(2^x + 2x\right) \mathrm{d}x$

(ii) $\int_2^{3.5} \left(2^{x+1} - 4x\right) \mathrm{d}x$

(3)

(Total for Question 6 is 9 marks)

(Total for Question 6 is 9 marks)

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Worked Solution - Question 6

Integration

Topic group

1. Set up the integral

Choose the correct integrand and limits, or apply the trapezium rule if the question gives tabulated values.

2. Evaluate the area or expression

Integrate or calculate the trapezium estimate carefully, then simplify the requested value.

3. State the final result

Write the answer in the form requested by the question: - = \times =

Final answer

- = \times =

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6 marks

Question 7

Circles

7. The circle C_1 has equation

$$x^2 + y^2 + 8x - 10y = 29$$

(a) (i) Find the coordinates of the centre of C_1

(ii) Find the exact value of the radius of C_1

(3)

In part (b) you must show detailed reasoning.

The circle C_2 has equation

$$(x - 5)^2 + (y + 8)^2 = 52$$

(b) Prove that the circles C_1 and C_2 neither touch nor intersect.

(3)

(Total for Question 7 is 6 marks)

(Total for Question 7 is 6 marks)

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Worked Solution - Question 7

Circles

Topic group

1. Identify the circle information

Use the centre-radius form, distance formula, gradient condition, or tangent property required by the diagram.

2. Complete the geometry

Substitute the coordinates carefully and simplify to the requested equation, length, point, or proof.

3. State the final result

Write the answer in the form requested by the question: Concludes as sum of radii < distance between centres, result proven * *

Final answer

Concludes as sum of radii < distance between centres, result proven * *

Question 8

12 marks

Trigonometric Equations

8. **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

(i) Solve, for $0 < x \leqslant \pi$, the equation

$$5\sin x \tan x + 13 = \cos x$$

giving your answer in radians to 3 significant figures.

(5)

(ii) The temperature inside a greenhouse is monitored on one particular day.

The temperature, H°C, inside the greenhouse, t hours after midnight, is modelled by the equation

$$H = 10 + 12\sin(kt + 18)° \qquad 0 \leqslant t < 24$$

where k is a constant.

Use the equation of the model to answer parts (a) to (c).

Given that

- the temperature inside the greenhouse was 20 °C at 6 am
- $0 < k < 20$

(a) find all possible values for k, giving each answer to 2 decimal places.

(4)

Given further that $0 < k < 10$

(b) find the maximum temperature inside the greenhouse,

(1)

(c) find the time of day at which this maximum temperature occurs.

Give your answer to the nearest minute.

(2)

(Total for Question 8 is 12 marks)

(Total for Question 8 is 12 marks)

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Worked Solution - Question 8

Trigonometric Equations

Topic group

1. Rearrange the trigonometric equation

Use the identity or ratio needed to reduce the equation to one trigonometric function in the required interval.

2. List the valid solutions

Find all angles in the given range and discard any extra values outside the interval.

3. State the final result

Write the answer in the form requested by the question: -= Time of day = 11:14

Final answer

-= Time of day = 11:14

Question 9

8 marks

Applications of Differentiation

9.

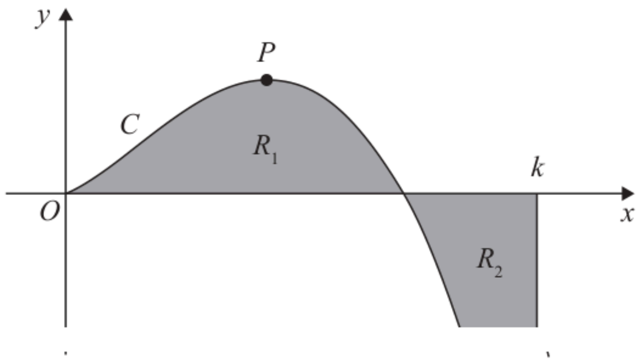

Figure 1

Figure 1 is a sketch of the curve C with equation

$$y = 2x^{\frac{3}{2}}(4 - x) \qquad x \geqslant 0$$

The point P is the stationary point of C.

(a) Find, using calculus, the x coordinate of P.

(4)

The region R_1, shown shaded in Figure 1, is bounded by C and the x-axis.

The region R_2, also shown shaded in Figure 1, is bounded by C, the x-axis and the line with equation $x = k$, where k is a constant.

Given that the area of R_1 is equal to the area of R_2

(b) find, using calculus, the exact value of k.

(4)

(Total for Question 9 is 8 marks)

(Total for Question 9 is 8 marks)

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Worked Solution - Question 9

Applications of Differentiation

Topic group

1. Differentiate and set the condition

Differentiate the expression, then use the gradient, stationary-point, tangent, normal, or optimisation condition.

2. Classify or evaluate

Substitute the required value and use the derivative information to complete the coordinate, gradient, interval, or maximum/minimum.

3. State the final result

Write the answer in the form requested by the question: The required value or expression is obtained by the final simplification.

Final answer

The required value or expression is obtained by the final simplification.

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Question 10

8 marks

Geometric Sequences

10. **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

The number of dormice and the number of voles on an island are being monitored.

Initially there are 2000 dormice on the island.

A model predicts that the number of dormice will increase by 3% each year, so that the numbers of dormice on the island at the end of each year form a geometric sequence.

(a) Find, according to the model, the number of dormice on the island 6 years after monitoring began. Give your answer to 3 significant figures.

(2)

The number of voles on the island is being monitored over the same period of time.

Given that

- 4 years after monitoring began there were 3690 voles on the island
- 7 years after monitoring began there were 3470 voles on the island
- the number of voles on the island at the end of each year is modelled as a geometric sequence

(b) find the equation of this model in the form

$$N = ab^t$$

where N is the number of voles, t years after monitoring began and a and b are constants. Give the value of a and the value of b to 2 significant figures.

(3)

When $t = T$, the number of dormice on the island is equal to the number of voles on the island.

(c) Find, according to the models, the value of T, giving your answer to one decimal place.

(3)

(Total for Question 10 is 8 marks)

(Total for Question 10 is 8 marks)

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Worked Solution - Question 10

Geometric Sequences

Topic group

1. Use the geometric formula

Identify the first term, common ratio, and whether the question needs a term, finite sum, or sum to infinity.

2. Substitute and solve

Apply the correct geometric formula and simplify, checking any validity condition on the ratio.

3. State the final result

Write the answer in the form requested by the question: - = 13

Final answer

- = 13